KRISHNA GUPTA

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SKILLS

Python • C/C++ • Surgical, AGV & AMRs Humanoid Robots • Teleoperation & Motion Retargeting (GMR) • Whole-Body Control (NVIDIA SONIC) • Imitation & Reinforcement Learning (Stable-Baselines3, RSL-RL, GR00T) • Simulation (Isaac Sim/Gym/Lab, MuJoCo) • Computer Vision & Deep Learning • LiDAR SLAM & Autonomous Navigation (ROS2, NAV2, FAST-LIO2, PX4/Ardupilot) • Edge AI (Jetson AGX Orin, TensorRT) • Real-Time Systems (Cyclone DDS, RTOS) • Embedded Firmware (STM32, ESP32, ARM Cortex-M) • BLDC/DC Motor Control (PID, CAN, Modbus)

EXPERIENCE

SS Innovations Private Limited, Gurugram — Surgical Robotics

Mechatronics Engineer, Robotics R&D | *July 2025 – Present*

- Member of the Robotics R&D team, working across the company's surgical robotics portfolio: the flagship SSi Mantra 3 multi-arm surgical teleoperation system and Kayaa, the in-house Unitree G1 humanoid surgical-assistant program.
- Architected Kayaa's cognitive architecture, integrating LLM and RAG pipelines for context-aware surgical assistance, while engineering and deploying an end-to-end TTS, LLM, and STT conversational AI stack directly on the Jetson AGX Orin.
- Engineered a Friction Compensation & Vibration Monitoring unit for the BLDC joint actuators at the core of the Mantra arm — real-time vibration capture, automatic load setting and analysis with closed-loop compensation.
- Architected Kayaa's VR teleoperation stack on the 29-DoF G1: Quest 3 controller/hand poses fused with ZED BODY_38 skeletons, GMR retargeting into NVIDIA SONIC whole-body control — 0.7 cm wrist accuracy, live-validated on hardware.
- Deployed the stack on Jetson AGX Orin; Developed a stereo vision return path (ZED → WebRTC) to Meta Quest.
- Own the robot-learning stack: imitation-learning data pipeline for GR00T N1.7 VLA action-space fine-tuning on New Embodiment and baseline locomotion policies trained across Isaac Sim / Isaac Gym and MuJoCo.
- Establishing simulation benchmarks and sim-to-real pathways for autonomous surgical subtasks—including basic suturing and ultrasound scanning—by adapting NVIDIA's Isaac for Healthcare workflows (Isaac Lab, RSL-RL, GR00T) to the Mantra 3 platform.

RigBetel Labs, Pune

Robotics Engineer (Firmware) | *December 2024 – June 2025*

- Integrated PX4 for UGVs with GPS-based autonomous navigation and IMU/GPS/vision sensor fusion; shipped a full ROS2 + NAV2 autonomous navigation stack for AMRs tied into real-time embedded firmware.
- Wrote low- and high-level ROS drivers (SPI, I2C, UART, CAN) for LiDARs, IMUs, and motor controllers; built a C++ motor controller with a low-level PID loop, Kalman-filter IMU+encoder odometry, and a GUI for live tuning.
- Enhanced firmware reliability through optimized ISRs under RTOS, hardware-mocking unit tests, and the development of an SSH-triggered OTA update pipeline for remote deployment.

Junior Embedded Engineer | *May 2024 – November 2024*

- Developed firmware across robotic platforms (Diadem, Acrux, Turtlebot Promax); built a robotic arm "Record & Play" mode using the drive servos themselves as motion-capture sensors — no added encoders; migrated legacy firmware to Micro-ROS.

Centre For Advanced Studies, AKTU, Lucknow

Mechatronics Research Intern | *June 2022 – December 2023*

- Research under Dr. Anuj Kumar Sharma (HOD, Mechatronics): built a colour-light communication system for miniature swarm robots using XGBoost for on-device colour recognition, plus a control/monitoring GUI with OTA updates over MQTT.
- Implemented gesture-based multi-robot control and a socket-based live terrain-monitoring pipeline; worked hands-on with Turtlebot3 and PX100 LoCoBot platforms.

iDEX - DIO, Ministry of Defence • Zebu Intelligent Systems Pvt Ltd, Hyderabad

Software Intern (Remote) *| May 2023 – July 2023*

- Built a real-time ship detection model (YOLOv7) robust to varied lighting, with low-light image restoration and synthetic haze augmentation for maritime safety.

PROJECTS / RESEARCH PAPERS

- “Carter”, Autonomous 4WD LiDAR Navigation Robot: Two variants of an autonomous mobile platform — an ESP32 version streaming LiDAR over UDP for off-board processing, and a Raspberry Pi 5 version with onboard navigation stack, autonomous docking, and basic LLM integration.
- “Vanguard”, low-cost robotic vehicle (eYRC 2023-24): Led the team building a visual-servoing system — overhead-camera event identification, path planning, and autonomous steering of a two-wheeled ESP32 robot using OpenCV.
- XGBoost-Enhanced Color Light Communication for Swarm Robotics (Springer, Arabian Journal for Science and Engineering): 96.66% classification accuracy at 0.403 ms execution on a 32-bit MCU (402 KB footprint), demonstrated across multiple swarming behaviours.

ACHIEVEMENTS

- **GATE 2024 ECE (Qualified) — 400 GATE Score**
- **e-Yantra Robotics Competition (eYRC) 2023-24, IIT Bombay — Top 30 in India**
  Designed the visual-servoing system for autonomous navigation.
- **ARTPARK Top-Up Fellowship, 2023-2024**
  Selected for support of innovative projects in Robotics, Autonomous Systems and AI.
- **XR Startup Grand Challenge (Meta • MeitY Startup Hub • FITT-IIT Delhi) — Top 20 in India**
  “Project Halia”, an XR personal-therapy assistant, shortlisted in the healthcare track.

EDUCATION

IET, DSMNRU, Lucknow • B.Tech (ECE) • (2020-2024) Percentage: 74.5%

La Martiniere College, Lucknow • ISC • (2020) Percentage: 81%